Supplementary Table S4

Graph Neural Networks for PM2.5 Forecasting: Review on Recent Advancements, Challenges, and Outlooks

Table S4: Complete characteristics of the 69 studies included in the systematic review: study identifier, publication year, study region, dataset or source, number of monitoring stations or graph nodes, forecast target, sampling interval, and forecasting horizon. All values are preserved from the study-level extraction; mixed or incompletely reported settings are stated as such.

Study	Year	Country	Dataset/source	Stations/nodes	Target	Sampling	Horizon
S01 Wang et al. (2023)	2023	China	Hourly air-quality observations from BJMEMC; meteorological observations from NCDC/NOAA	Air-quality stations = 17; meteorological stations = 9; PM2.5 graph nodes = 17; meteorological graph nodes = 9	PM2.5	1 h	1–72 h
S02 Zhou et al. (2024)	2024	China	Beijing air-quality dataset; Tianjin air-quality dataset; Wuhan air-quality dataset	Beijing air-quality stations = 35; Beijing meteorological stations = 18; Tianjin stations = 7; Wuhan stations = 21	PM2.5 (Beijing); AQI (Tianjin); PM2.5 (Wuhan)	1 h	12 h; 18 h; 24 h
S03 Huo et al. (2026)	2026	China	Four publicly available city-level air-pollutant datasets from the National Urban Air Quality Real-time Release Platform	Graph nodes are cities: Guangdong = 21; Hubei = 11; Liaoning = 14; Fenwei Plain = 10	PM2.5; PM10; NO2	1 h	Next 1 sampling interval
S04 Jana et al. (2024)	2024	India; United States	Delhi continuous air-quality and meteorological monitoring data; California air-quality monitoring data	Delhi: 40 monitoring stations total; usable by variable — PM2.5 = 28, PM10 = 27, NO2 = 27, CO = 29, SO2 = 31, O3 = 31, meteorology = 30. California: about 240 stations total; usable PM2.5 = 75, NO2 = 63, CO = 43.	PM2.5; PM10; NO2; O3; CO; SO2 (Delhi); PM2.5; NO2; CO (California)	1 h	1 h; 24 h; 48 h
S05 Qi et al. (2019)	2019	China	Ground air-quality monitoring data and meteorological observations for the Jing-Jin-Ji region	76 air-quality monitoring stations	PM2.5	1 h	1 h; 2 h; 4 h; 8 h; 12 h; 24 h; 48 h; 72 h
S06 Xiao et al. (2022)	2022	China	MEE city-level PM2.5 observations; ERA5 meteorological reanalysis	184 city-level observation stations/nodes	PM2.5	3 h	3, 6, 12, and 24 time steps; corresponding reported horizons = 9 h, 18 h, 36 h, 72 h
S07 Sui and Han (2023)	2023	China	Beijing real-world air-quality/meteorological/POI dataset; Tianjin real-world air-quality/meteorological/POI dataset	Beijing stations/nodes = 35; Tianjin stations/nodes = 26	PM2.5 (continuous value) with auxiliary PM2.5 pollution-level classification	1 h	Next 6 h
S08 Teng et al. (2023)	2023	China	Ground air-quality monitoring; meteorological observations; MERRA-2 AOD; SRTM elevation	Original central Beijing sites = 12; expanded graph/site coverage = 140 sites in total	PM2.5	1 h	1–72 h; reported evaluations include 6 h; 12 h; 24 h; 48 h; 72 h
S09 Zhou et al. (2021)	2021	China	National air-quality monitoring PM2.5 observations; CERA-SAT wind-field reanalysis	57 selected monitoring stations/nodes (from 108 national-control stations)	PM2.5	1 h	1 h; 2 h; 4 h; 8 h; 12 h; 24 h; 36 h; 48 h
S10 Zhou et al. (2022)	2022	China	National real-time urban air-quality observations; ERA5-Land meteorological data	71 monitoring stations/nodes	PM2.5	1 h	1 h; 2 h; 4 h; 8 h; 12 h; 24 h; 48 h; 72 h
S11 Chen et al. (2025)	2025	China	CNEMC air-quality observations; NOAA historical meteorology; NCEP GFS forecast meteorology	29 city nodes	PM2.5	Air quality = 1 h; historical meteorology = 3 h then interpolated hourly; GFS forecast meteorology = 3 h	3–48 h in 3-h increments
S12 Guyu et al. (2025)	2025	China	CNEMC air-quality monitoring data; ERA5/Copernicus meteorological data	86 monitoring stations/nodes	PM2.5	1 h	1–48 h multi-step; detailed comparisons include 36 h and 48 h
S13 Shi et al. (2026)	2026	China; India	UCI Beijing Multi-Site Air-Quality Data; Kaggle Air Quality Dataset Indian Cities 2022–2025	Beijing = 12 monitoring stations/nodes; India = 8 city nodes (station count NR)	PM2.5	1 h	1 h; 6 h; 12 h; 24 h
S14 Zhao et al. (2026)	2026	China	CNEMC National Urban Air Quality Real-Time Release Platform; Copernicus/ERA5 meteorological data	Main dataset = 114 monitoring stations/nodes; additional comparison dataset = 86 stations	PM2.5	1 h	1–12 h (tables report +1 through +12 h)
S15 Ni et al. (2023)	2023	China	KnowAir; Microsoft UrbanAir (North subset)	KnowAir = 184 regions/nodes; UrbanAir(North) = 181 total regions, 125 accepted/used nodes	PM2.5	KnowAir = 3 h; UrbanAir(North) = 1 h	1 time step; 8 time steps
S16 Kim et al. (2023)	2023	South Korea; China (auxiliary upwind-city PM2.5 information)	Airkorea PM2.5 monitoring; ECMWF/reanalysis meteorology and atmospheric variables; land-cover and population information; auxiliary China PM2.5	101 South Korean monitoring stations/nodes	PM2.5	3 h	3 h; 6 h; 9 h; 12 h; 24 h; 48 h; 72 h; 96 h
S17 Zeng et al. (2025)	2025	China	Beijing public air-quality/meteorological dataset; Chongqing CNEMC dataset; Guangzhou and Chengdu datasets for transfer/generalization experiments	Beijing = 12 sites/nodes; Chongqing = 28 sites/nodes; Guangzhou/Chengdu node counts = NR in main text	PM2.5 (main); additional experiments predict PM10; NO2; O3	1 h	1–6 h; 1–12 h; 1–24 h
S18 Liu et al. (2026)	2026	China	CNEMC air-quality data; NCEI Integrated Surface Database meteorology	Monitoring stations: Beijing = 3; Chengdu = 4; Changchun = 3. Proposed S-DI graph nodes represent latent dimensions/components rather than monitoring stations.	PM2.5	1 h	1 h; 3 h; 6 h; 12 h
S19 Xu et al. (2023)	2023	China	MEE city-level PM2.5 observations; ERA5 meteorological reanalysis	184 city-level observation stations/nodes	PM2.5	3 h	3, 6, 12, and 24 time steps; reported equivalents = 9 h, 18 h, 36 h, 72 h
S20 Ma et al. (2026)	2026	China; India	Beijing 2013–2017: UCI Beijing Multi-Site Air-Quality Data; Beijing 2020–2023: National AQ + ISD-Lite; Chengdu 2020–2023: National AQ + ISD-Lite; Delhi 2017–2020: Delhi AQ + NOAA ISD-Lite	Beijing 2013–2017 = 12; Beijing 2020–2023 = 12; Chengdu = 8; Delhi = 12; graph nodes = monitoring stations	PM2.5	1 h	1 h; 6 h; 12 h; 24 h
S21 Su et al. (2023)	2023	Taiwan	EPA air-quality monitoring data; Central Weather Bureau meteorological observations; AirBox micro-sensor data for non-monitoring-location evaluation	Air-quality monitoring stations/graph nodes = 84; AirBox target locations used for non-monitoring-area evaluation = 19	PM2.5	1 h	Next 48 h
S22 Zhao et al. (2025)	2025	China	CNEMC air-quality monitoring data	Beijing = 35; Tianjin = 21; Henan = 98 monitoring stations/nodes	PM2.5	1 h	NR
S23 Gao et al. (2026)	2026	China	2023 China National Air Quality Dataset; 2019 Yangtze River Delta Air Quality Dataset	Main dataset: 375 initial stations, 342 retained monitoring stations/nodes after missing-data filtering; Yangtze River Delta: 142 initial stations, 117 retained monitoring sites/nodes	PM2.5	1 h	Next 1 h

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Supplementary Table S4 (continued)

Study	Year	Country	Dataset/source	Stations/nodes	Target	Sampling	Horizon
S24 Zhang et al. (2024)	2024	China	KnowAir dataset; PM2.5 observations from the Ministry of Ecology and Environment; meteorological variables from the Global Forecast System	184 cities/nodes	PM2.5	3 h	3 h; 6 h; 12 h; 24 h; 36 h; 48 h; 60 h; 72 h
S25 Mandal and Thakur (2023)	2023	India	CPCB ground air-quality monitoring-station data; ERA5/ECMWF meteorological data	38 air-quality monitoring stations/nodes	PM2.5	3 h	3 h; 6 h; 9 h; 12 h; 15 h; 18 h; 21 h; 24 h; additional 72 h experiment
S26 Hu et al. (2024)	2024	China	Beijing Environmental Protection Monitoring Center air-quality data; ECMWF/Copernicus meteorological data	39 monitoring stations/nodes	PM2.5	1 h	1 h; 3 h; 6 h; 12 h; 24 h
S27 Tan et al. (2022)	2022	China	Research monitoring data collected at six urban sites using an American TSI environmental measuring instrument	6 monitoring sites/nodes; Sites 1–3 are the principal prediction datasets and Sites 4–6 provide spatial information	PM2.5	5 min	NR
S28 Gao and Li (2021)	2021	China	China Environmental Monitoring Center hourly air-quality data with meteorological observations from the nearest meteorological station to each air-quality station	32 air-quality monitoring stations/nodes; meteorological-station count NR	PM2.5	1 h	1 h; 2 h; 4 h; 8 h; 12 h; 24 h; 48 h
S29 Liu et al. (2021)	2021	China	Research-collected multi-site urban PM2.5/pollutant monitoring data from six sites	Monitoring sites/nodes = 6; Sites 1–3 are principal target/evaluation datasets and Sites 4–6 provide spatial information	PM2.5	5 min	NR
S30 Zhang et al. (2024)	2024	China	NWNU-FDAir (Northwest Normal University-Four Districts' Air) dataset from dense urban micro-monitoring stations	Monitoring stations/graph nodes = 456 micro-monitoring stations	PM2.5	1 h	T+1 h; T+2 h; T+3 h; T+4 h; T+5 h; T+6 h; T+7 h; T+8 h
S31 Ma et al. (2022)	2022	China	Public heterogeneous Beijing air-quality and meteorological dataset from Harvard Dataverse (Wang, 2019)	Meteorological stations=35; air-quality stations=17; paper also states total monitoring points/graph nodes=53 and uses a 53×53 graph — Unclear from full text because 35+17=52	PM2.5	1 h	3 h
S32 Gu et al. (2025)	2025	China	CNEMC pollutant monitoring data combined with meteorological-station observations	Pollutant monitoring stations=37; meteorological stations=9; subgraphs are constructed per target station with a maximum of 9 pollutant stations per cluster/subgraph before integrating relevant meteorological stations	PM2.5	Pollutant data=1 h; meteorological data have station-dependent 1 h, 3 h or 6 h intervals	1–72 h multi-step; main comparative experiment reports 24 h and extended prediction intervals 1–24 h, 25–48 h, and 49–72 h
S33 Zhang et al. (2024)	2024	China	Historical air-pollutant monitoring data from Beijing Municipal Environmental Protection Monitoring Center/National Urban Air Quality Real-Time Monitoring System; meteorological data from U.S. National Climatic Data Center	57 pollutant monitoring stations/nodes	PM2.5	1 h	1 h; 8 h; 12 h; 24 h; 48 h; 72 h
S34 Zhao et al. (2025)	2025	China	Hourly CNEMC PM2.5 observations; ERA5 10-m wind reanalysis	200 stations in study area; 50 stations with <5% missing data selected as experimental graph nodes	PM2.5	1 h	6 h; 12 h; 18 h; 24 h; 30 h; 36 h
S35 Zhang et al. (2025)	2025	China	CNEMC air-quality data combined with ECMWF meteorological/satellite-related environmental data; dataset/code repository supplied by authors	Graph nodes are cities: YRD=18; BTH=10; PRD=9	PM2.5	1 h	24 h; 48 h; 72 h
S36 Ye et al. (2025)	2025	China	PM2.5 observations from Ministry of Ecology and Environment; ERA5 meteorological and geographic data	Initial spatial units/nodes = 40 cities; IPS filtering retains a target-specific subset of surrounding cities, so the effective graph node count varies by target	PM2.5	3 h	3 h; 6 h; 9 h; 15 h; 24 h
S37 Velez-Serrano and Alvaro-Meca (2026)	2026	Spain	CAMS air-pollution reanalysis; ERA5-Land meteorology; elevation; municipal population; public-holiday calendar; author-provided OSF dataset	Municipal graph nodes=8,076; spatial extrapolation is additionally produced on a 0.1°×0.1° grid	PM2.5	Daily (hourly source data are aggregated to daily values for analysis)	1 day ahead (H=1)
S38 Dong et al. (2024)	2024	China	Two real-world air-quality datasets from a public air-quality monitoring website plus meteorological data website; exact URLs supplied in the paper	Dataset 1=21 stations/nodes; Dataset 2=36 stations/nodes	Air-quality/pollutant concentration sequence; the supplied full text lists six pollutants but does not identify one single primary pollutant target for all AirNN experiments — Unclear from full text	Dataset 1=3 h; Dataset 2=1 h. Dataset 1 is nevertheless evaluated with a reported 1-h forecast column — Unclear from full text.	1 h; 6 h; 12 h; 24 h
S39 Dun et al. (2022)	2022	China	Beijing PM2.5 Dataset; Fushun PM2.5 Dataset	Beijing = 12 monitoring stations/nodes; Fushun = 34 monitoring stations/nodes	PM2.5	1 h	1 h; 2 h; 3 h; 4 h
S40 Tariq et al. (2023)	2023	South Korea	Air-quality monitoring data from selected stations in Gyeonggi-do Province	23 monitoring stations total; 18 known/training locations; 5 held-out unknown locations for remote-sensing evaluation	PM2.5; PM10; NO2; O3; CO	1 h	1 h; 3 h; 6 h; 12 h
S41 Wang et al. (2024)	2024	China	Hourly air-quality data for 56 cities; ERA5 meteorological data	Model-training graph = 56 city nodes; core collaborative-control study area = 28 "2+26" cities	PM2.5; O3	1 h	1 h; 4 h; 8 h; 24 h; 48 h; 72 h; 96 h; 120 h; Avg8 h; Avg24 h; Avg48 h; Avg72 h
S42 Zhang et al. (2024)	2024	China	PM2.5 observations from the Ministry of Ecology and Environment; meteorological/environmental data from ERA5	184 city nodes; 3,938 graph edges	PM2.5	3 h	3 h increments over 24 steps up to 72 h (3, 6, 9, 12, 15, 18, 21, 24, 27, 30, 33, 36, 39, 42, 45, 48, 51, 54, 57, 60, 63, 66, 69, 72 h)
S43 Liu et al. (2024)	2024	China	PM2.5 Forecasting Benchmark released by the authors	Original stations=534; retained graph nodes after removing stations with >30% missing data=415	PM2.5	1 h	24 h ahead
S44 Mohammadzadeh et al. (2024)	2024	United States	U.S. EPA PM2.5 AQI data; Weather Underground meteorological observations	13 active EPA sites initially identified; 7 excluded for data gaps; final graph nodes/sites=6	PM2.5 AQI	Daily after preprocessing	Candidate historical/prediction configurations from 1–5 days were examined; final model uses 3 days of history to forecast the following day
S45 Liao et al. (2024)	2024	United States	CARB and U.S. EPA PM2.5 observations; ERA5 meteorology; VIIRS Suomi fire radiative power; CAL FIRE prescribed-burn records	112 PM2.5 monitoring locations/nodes	PM2.5	1 h	1–48 h prediction window
S46 Han et al. (2025)	2025	China	One year of air-pollutant data used in the study; exact named dataset/source NR	Monitoring stations are referenced, but the number of stations/graph nodes is NR	PM2.5; NO2; SO2	NR	NR
S47 Huang et al. (2025)	2025	China	PM2.5 dataset described as the Beijing-Tianjin-Hebei region dataset; exact source agency NR	Major cities/stations are selected; exact number of graph nodes NR	PM2.5	3 h	3 h; 6 h; 9 h; 18 h; 36 h; 72 h (reported as steps 1, 2, 3, 6, 12 and 24 at 3-h sampling)
S48 Wang et al. (2024)	2024	China	China National Environmental Monitoring Centre air-quality data; ERA5 meteorological data; Gaode e-map API POI data	8 state-controlled air-quality monitoring stations/nodes	PM2.5; PM10; NO2; O3; SO2; CO	1 h	1 h; 3 h; 6 h; 12 h; 24 h

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Supplementary Table S4 (continued)

Study	Year	Country	Dataset/source	Stations/nodes	Target	Sampling	Horizon
S49 Dimri et al. (2024)	2024	South Korea	AirKorea surface observations; WRF v3.8 meteorological forecasts; CMAQ v5.2 pollutant forecasts	183 AirKorea monitoring stations/nodes	PM2.5	1 h	+1 h; +3 h; +6 h; +12 h; +24 h; +48 h; +72 h
S50 Qi et al. (2024)	2024	China	Seven years of pollutant-concentration and meteorological data from 30 northern Chinese cities; exact named data provider NR	Main graph nodes = 30 cities; additional Shenyang fine-scale experiment = 72 monitoring sites	PM2.5	NR (window length L and cycle T are expressed in hours)	NR
S51 Pan et al. (2024)	2024	China	Pollutant data from Ministry of Environmental Protection of China / China National Environmental Monitoring Center; meteorological data from ECMWF	308 city nodes	PM2.5	3 h	Next 72 h (24 steps at 3-h sampling); experiments report T1=24 with T0=1 and T0=24 history settings
S52 Jin et al. (2023)	2023	China	Beijing-PM2.5-Concentration-Data public dataset	16 district nodes	PM2.5	1 h	Next 1 h (24 historical hourly points predict the 25th point)
S53 Zhang et al. (2022)	2022	China	Beijing PM2.5 dataset; Taizhou PM2.5 dataset collected by the authors' team	Monitoring stations: Beijing dataset = one city/station source as used in the paper; Taizhou = single Hongjia monitoring station. Graph nodes represent time moments constructed from a univariate/multivariate time-series window; exact graph-node count NR.	PM2.5	1 h	1 h; 6 h; 12 h; 24 h; 48 h
S54 Zeng et al. (2022)	2022	China	Real atmospheric dataset collected in home-based care parks	9 monitoring stations/nodes	PM2.5	10 min	Next 10 min
S55 Lu et al. (2025)	2025	China	National Urban Air Quality Real-Time Release Platform	22 monitoring stations/nodes; model additionally learns 3 station groups in the selected configuration	PM2.5	1 h	16 h; 24 h; 32 h
S56 Abbasi et al. (2025)	2025	Iran	Air-quality data from Tehran Municipality and Department of Environment; meteorological observations from Iran Meteorological Organization	11 air-quality monitoring stations/nodes; meteorological-station count NR	PM2.5	1 h	1 h; 2 h; 4 h; 8 h; 12 h; 24 h; 48 h; 72 h
S57 Guan et al. (2025)	2025	China	CHAP air-pollutant dataset; ERA5-Land meteorology; AOD-LGHAP from National Earth System Science Data Centre; NASA NEO NDVI	173 raster grid cells/nodes at 10 km spatial resolution	PM2.5	Daily	First step; second step; third step (daily multi-step forecasting; 1, 2, and 3 days)
S58 Chen et al. (2024)	2024	China	Hourly Beijing pollutant concentration data; corresponding meteorological data; MapWorld API POI data	Prediction target stations = Fangshan; Tiantan; Dongsi (3 target stations). Total monitoring stations/graph nodes = NR	PM2.5; PM10; O3	1 h	7-step multi-step forecasting (up to 7 h)
S59 Chauhan et al. (2026)	2026	China	UCI Beijing Multi-Site Air Quality Database	12 monitoring stations/nodes; 420,768 data points	PM2.5	1 h	Next time step (1 h)
S60 Li et al. (2023)	2023	China	Hourly air-quality data from 12 nationally controlled monitoring sites; meteorological data from China Meteorological Administration	12 air-quality monitoring stations/nodes	PM2.5; PM10; SO2; NO2; CO; O3	1 h	1 h; 8 h; 16 h; 24 h
S61 Mirzaei and Torghabeh (2026)	2026	China	UCI Beijing Multi-Site Air-Quality Data; processed hourly subset from five stations	5 monitoring stations/nodes; 720 hourly records per station	PM2.5; PM10; NO2; SO2	1 h	72 h
S62 BADOUCH and BELHOUCINE (2026)	2026	China	Public Beijing Multi-Site Air Quality Dataset (UCI #501)	12 stations/nodes	PM2.5	1 h	1 h
S63 Huang et al. (2025)	2025	China	PM2.5 observations from the Ministry of Ecology and Environment of China; ERA5 atmospheric reanalysis meteorological/environmental indicators	184 city nodes	PM2.5	3 h	3 h; 6 h; 12 h; 24 h; 36 h; 48 h; 60 h; 72 h
S64 Pei et al. (2022)	2022	China	China Environmental Monitoring Station air-quality monitoring dataset	The paper averages values across all monitoring sites to form the Beijing series; number of monitoring stations = NR. Graph nodes represent multivariate time-series variables; exact graph-node count = NR.	PM2.5	1 h	1–12 h; comparison table explicitly reports +1 h, +2 h, +3 h; ablation reports average 12-h prediction
S65 Yu et al. (2026)	2026	China	Raw air-quality and meteorological data shared by the authors in the study repository	76 air-pollution monitoring stations/nodes	PM2.5	1 h	1 h; 2 h; 4 h; 6 h; 8 h; 10 h; 12 h
S66 Wang et al. (2024)	2024	China	Air-quality monitoring data from 30 Beijing monitoring stations; source website reported as bjmemc.com.cn	30 monitoring stations/nodes	PM2.5; PM10; CO; O3; NO2 (five-pollutant generalization experiment)	1 h	1–24 h, reported in groups 1–6 h; 7–12 h; 13–18 h; 19–24 h. Long-term table reports 25–30 h; 31–36 h; 37–42 h; 43–48 h. Generalization experiment evaluates 1–700 h.
S67 Zhao (2025)	2025	China	CNEMC air-quality monitoring data; ERA5 meteorological data; NDVI data; DEM data	Air-quality monitoring stations/nodes = 65; meteorological stations = 17	PM2.5	Daily averages derived from hourly pollutant data	Annual-average PM2.5 predictions for 2025–2030; conventional short-horizon forecast interval NR
S68 Zeng et al. (2021)	2021	China	Real atmospheric monitoring dataset from Dongmingnan Industrial Park	9 monitoring stations/nodes	PM2.5	10 min	Next 10 min
S69 Wang et al. (2022)	2022	China	KnowAir public air-pollution dataset	184 city nodes	PM2.5	3 h	72 h (24 time steps)

Notes. Values reproduce the supplied 69-study extraction. NR = not reported; uncertain or partially described properties are not inferred. S01–S69 are row identifiers (full titles and DOIs in the companion study registry). Locations and datasets may contain multiple regions or monitoring networks. A forecast in steps is not converted into hours without explicit sampling information.